



Figure 4: Block editor image 2

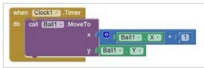


Figure 5: Block editor image 3

discuss the actual functionality that we are expecting from our application.

1. We need to code it so that on pressing the *Start* button, the ball starts moving.
2. When the *Stop* button is pressed, it should stop at its current location.
3. On pressing the *Fast* button, its speed should increase.
4. On pressing the *Slow* button, its speed should decrease.

So let's move on and add these functions using the block editor. I hope you remember how to switch from the designer to the block editor. There is a button available right above the *Properties* pane to switch between them.

## Block editor blocks

I have already prepared the blocks for you (Figures 3 to 7). All you need to do is to drag the relevant blocks from the left side palette and drop them on the viewer. Arrange the blocks the same way you see them in the image. I will explain what each does and how it is called.

1. The blocks in the figures are easy to understand. The very first block tells us that when the *Start* button is pressed, it should enable only the clock in the application.
2. When the timer is enabled, it will increase the x coordinate of the ball, which will move it towards the right, keeping the y coordinate as it is. Here, 'plus 1' means that on every cycle of the clock component, it will increase the horizontal position of the ball by one count.
3. When the *Fast* button is pressed, it will increase the speed component



Figure 6: Block editor image 4

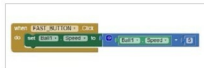


Figure 7: Block editor image 5

- of the ball, causing it to move faster.
4. When the *Slow* button is pressed, it will decrease the speed component of the ball, causing it to move slower.
  5. When the ball has reached the edge, it should start this all over again. Hence, for this we need to set the x coordinate as 0.

With this, the block editor is done with. Next, we will move to downloading and installing the app on your phone to check how it is working.

## Packaging and testing

To test the app, you need to get it on your phone. First, you have to download the application to your computer and then move it to your phone via Bluetooth or USB cable. I'll tell you how to download it.

1. On the top row, click on the *Build* button. It will show you the option to download the *apk* to your computer.
2. Downloading will show the progress and after a successful download, the application will be placed in the download folder of your directory or the preferred location you have set for your app.
3. Now you need to get this *apk* file to your mobile phone either via Bluetooth or USB cable. Once you have placed the *apk* file in your SD card, you need to install it. Follow

the on-screen instructions to install it. You might get some notifications or warnings saying 'Install from untrusted source'. Allow this from the settings and after successful installation, you will see the icon of your application in the menu of your mobile. Here you will see the default icon, which can be changed. I will tell you more about this, as we move ahead in this course.

Your application should now work exactly as per your requirements. Depending upon your usability and customisation, you can change various things like images, sound and behaviour too.

## Debugging the application

We have just created the prototype of the application with very basic functionality. What else might users be interested in? Now come various use cases, which require serious analysis as not to annoy the user. Consider the following:

1. What are the other factors that will affect the movement of the ball?
2. Rather than buttons, can we make the ball move from the sensor commands (orientation sensor)?
3. How can we change the direction the ball moves in? In the example mentioned earlier, the ball moves towards the right; how can we make it move in the up-down axis?

These are some of the possible scenarios that users will be pretty happy seeing implemented in your app. So think over how these can be integrated into the application. Do ask me if you fail to accomplish any of the above cases.

Congratulations! You have successfully built another useful Android app for yourself. Happy inventing! 

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